

POANET: Parts-Based Occlusion Aware Network for Person Re-Identification

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Abstract—Occlusion remains a major challenge in person re-identification (ReID), as it corrupts feature representations and reduces retrieval accuracy. Most existing methods rely on proxy strategies such as random erasing or pose cues, but they lack explicit modeling of where and how occlusion occurs. To address this gap, we propose two key innovations. First, an Occlusion Detection Network (ODN) explicitly estimates the extent and location of occlusion, enabling the ReID backbone to downweight corrupted regions and emphasize features from visible regions. Second, a Part Aware Feature Extractor (PAFE) generates anatomically correct part embeddings (head, torso, legs), allowing the model to dynamically shift focus from occluded to unoccluded parts in a self-supervised manner while preserving global context. The paper also introduces SHHQ-occ, a large-scale dataset of 1.2M images with precise, part-level occlusion annotations, providing a robust foundation for occlusion modeling. Experiments on occluded benchmark datasets: Occluded-ReID, Occluded-Duke, P-DukeMTMC-ReID, PETHZ and holistic reid datasets: Market-1501, CUHK03, DukeMTMC, MSMT17 demonstrate that our approach achieves state-of-the-art accuracy with strong occlusion robustness and efficiency.

Index Terms—Human parsing, occlusion modeling, person re identification, super-resolution.

I. INTRODUCTION

PERSON Re-Identification (ReID) is a challenging task in computer vision, with its main goal being to identify a specific individual from a gallery of images across different cameras. Despite significant advancements in the field, with myriad techniques to incorporate occlusion robustness in diverse architectural choices, two fundamental shortcomings persist in current ReID methods: (i) the scarcity of approaches that explicitly and physiologically model occlusion at fine granularity with precise anatomically aware occlusion quantification, and (ii) methods that employ feature map division or image partitioning to achieve finer granularity in the model's feature extraction, while improving accuracy in certain scenarios, do not consider the semantic meaningfulness

of the parts they divide the image into, which can lead to misalignment between feature extraction and actual body parts in the image.

In this paper, we address these two key limitations through two novel contributions. First, we introduce a **Occlusion Detector Network (ODN)** that accurately identifies the location and extent of occlusions in the input images. This information is then used as a transfer learning mechanism in our ReID pipeline using our **Occlusion Awareness Module (OAM)**, where the model dynamically adjusts its attention to focus on unoccluded regions while suppressing attention on occluded regions. This self-supervised approach provides a more reliable feature representation, particularly in challenging occlusion scenarios. The second novelty is the introduction of the **Part Aware Feature Extractor (PAFE)**, a network that divides the image into semantically meaningful parts—head, torso, and legs—ensuring that the model maintains semantic information about the body parts it is processing. By using PAFE, we can focus on specific body parts in a way that aligns with their anatomical significance, improving the model's ability to learn discriminative features for person ReID.

In contrast to previous methods that rely on global feature adaptation [1] or indirect proxies for occlusion [2], our method provides *interpretable, direct information* about which body parts are occluded and to what degree. It allows occlusion information to be used in a targeted, part-wise manner for feature aggregation and comparison, improving both performance and interpretability. Our framework operates without the need for external pose, silhouette, or annotation cues. It autonomously partitions each person image into anatomically meaningful regions (head, torso, legs) and employs ODN to generate fine-grained occlusion maps for each part. PAFE and ODN work jointly: when ODN detects high occlusion in a part (e.g., legs), the model dynamically downweights the feature embedding of only that part, relying more heavily on unobstructed regions. This behavior is directly evidenced in our ablation studies and qualitative results.

To train our occlusion detector, we developed a simple yet elegant algorithm that generates a realistic occluded image dataset **SHHQ-occ**. By overlaying common occluding objects, such as benches, road signs, and bushes, onto full-body human images from the SHHQ dataset, we simulate real-world occlusions. We devised a method to generate precise ground-truth labels for the extent and location of occlusion in each image. This dataset ensures accurate occlusion labeling,

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providing our model with reliable data for learning to detect occlusion effectively.

We unify these innovations into a single retrieval pipeline, where the ODN in conjunction with the OAM guides the attention mechanism and PAFE ensures semantic coherence across the image. Extensive experiments on benchmark datasets, including Market-1501, CUHK03, and DukeMTMC, demonstrate that our approach achieves state-of-the-art performance across the board, outperforming existing methods in handling occlusions and ensuring fine-grained feature extraction. Importantly, we achieve these results while maintaining computational scalability by employing only convolutional neural networks, thereby avoiding complex attention mechanisms or expensive large covariance matrix calculations that we encounter in transformer based approaches.

In summary, our main contributions are:

- 1) We introduce a novel **Occlusion Detector** and an **Occlusion Awareness Module** that explicitly model occlusion in person ReID systems, without relying on external body part cues or data augmentation, while maintaining low computational overhead.
- 2) We propose the **Part Aware Feature Extractor (PAFE)** module, which generates semantically meaningful part embeddings. This module adds anatomical significance to the system and, in conjunction with the occlusion detector, dynamically adjusts focus to emphasize regions that contribute to more accurate retrieval.
- 3) We create a **comprehensive holistic dataset (SHHQ-occ)** consisting of 1.2 million images generated using the SHHQ dataset and a variety of occluding objects to simulate real-world occlusion. This dataset is used to train our occlusion detector and provides precise information regarding the extent and location of occlusion in human body images. Due to restrictions from the SHHQ dataset authors, we are unable to release the dataset directly. However, we provide a detailed algorithm and workflow for generating this dataset once the required data are available to interested researchers.
- 4) We consolidate above innovations into a **single unified ReID pipeline**. Our work fundamentally differs from prior OCC-ReID approaches by explicitly modeling fine-grained, anatomically structured occlusion maps within a self-contained ReID framework, without relying on external cues or manual annotations. Unlike previous methods—which often address occlusion implicitly or through indirect proxies—our Occlusion Detector Network and Occlusion Awareness Module provide direct, interpretable, and physiologically meaningful signals. Our method demonstrates state-of-the-art performance across eight benchmark datasets, showcasing the efficacy of our approach in addressing occlusion and fine-grained feature extraction.

II. RELATED WORK

A. Occlusion in Computer Vision

Modeling occlusion has long been studied in artificial intelligence, with prior efforts focusing on bounding

recognition errors or estimating distortion effects. Boshra and Bhanu [3] established upper bounds on recognition error under occlusion, while Wang et al. [21] proposed a distortion model that estimates optimal gallery size via constrained feature map replacement. Han and Bhanu [22] simulated gait recognition performance under varying within-class variance. More recent approaches address occlusion through modality- or task-specific strategies: Yang et al. [23] used SMPL edge feature distillation for pose estimation; Zhang et al. [18] combined RF-vision with cross-modality distillation for 3D hand pose; Fan et al. [19] introduced Occ²Net for 3D occupancy-based matching; Lee et al. [24] developed Latent-OFER for reconstructing occluded facial regions; Ke et al. [20] proposed bilayer networks to separate occluder and occludee; and Gambashidze et al. leveraged X-ray distillation to mitigate LiDAR occlusions.

Gupta and Chellappa [12] proposed a separate occlusion detector module to guide a gait recognition backbone; however, their detector is trained solely on silhouettes, limiting its ability to infer occlusion extent directly from RGB images and forcing its reliance on segmentation, which can be error-prone.

In contrast, our method explicitly integrates an **Occlusion Detection Network** with anatomically meaningful **Part-Aware Feature Extractor (PAFE)** embeddings, enabling robust, self-supervised focus shifting from occluded to visible regions. Unlike existing approaches that are either statistical, modality-specific, or reconstruction-based, ours is the first to provide a unified, general, and computationally efficient solution for person ReID under occlusion, achieving both state-of-the-art accuracy and practical scalability.

B. Person Re-Identification

Deep learning has significantly advanced ReID systems by moving beyond hand-crafted features. Early Siamese network models [25], [26], [27] surpassed traditional methods but lacked explicit mechanisms for handling occlusion, while IDE-based (Identity Discriminative Embedding) approaches [28] established robust baselines with global embeddings but overlooked the effects of partial occlusion. Mid-level feature models [29], [30] attempted to capture local semantic relations but showed limited robustness under partial visibility. More recent multi-channel [31] and part-based frameworks [32] improved local-global feature integration, yet still lacked explicit occlusion modeling, often leading to degraded results under heavy occlusion or pose variation. Chen et al. [14] and Yan et al. [33] incorporate attention and pose information to address occlusion, while Somers et al. [34] and Jonnalagedda and Bhanu [35], [36] generated representations of external body parts to compensate for missing regions. Gao et al. [37] introduced an occlusion awareness score within the network to localize occluded parts. However, these works lacked explicit information about the type and precise location of occlusion, which we demonstrate is valuable for guiding recognition models under heavy occlusion. Our approach, presented in this paper directly modeled and leverages this occlusion information, offering a distinct advantage over prior methods.

Landmark-based [15], [38] approaches further boosted part-level discrimination, albeit with increased dependence on accurate keypoint detection or pose cues. Meanwhile, transformer-based models [1], [6], [7], [8], [39] used attention heads to dynamically focus on image regions, but this came at a significant computational cost, requiring large-scale covariance calculations and considering pixel-level interactions, including irrelevant background pixels. More recently, Si et al. [40] proposed SDN to recover occluded regions using semantic-guided decoding, but its generative design adds complexity and inefficiency. Tan et al. [2] introduced SPT, leveraging semantic part annotations with transformer attention, though it depends on external annotations and incurs heavy computation. In contrast, our work explicitly models occlusion using lightweight convolutional modules, avoiding annotation dependence and ensuring scalability while preserving robustness under diverse occlusion scenarios.

III. SHHQ-OCC DATASET

We present a novel dataset specifically designed for precise, part-level occlusion modeling in human-centric vision tasks. While this dataset is central to our occlusion-aware person ReID framework, its utility extends to a wide range of occlusion-related problems in computer vision. Existing occlusion modeling efforts often rely on coarse or proxy labels, such as silhouette-based masks or binary occlusion indicators, which lack granularity and fail to capture nuanced occlusion patterns. In contrast, our dataset provides **exact, continuous-valued occlusion annotations** for three semantically meaningful regions of the human body—head, torso, and legs—represented as a tuple $(x, y, z) \in [0, 1]^3$, where each element quantifies the fraction of that region's area occluded.

This fine-grained representation enables direct application to person ReID under occlusion, where both supervised and regression-based occlusion estimation modules can use it to guide retrieval pipelines. It can further support diverse human-centric tasks by providing occlusion information for gait recognition, serving as ground truth for visibility prediction in pose estimation and tracking, improving instance segmentation through region-wise occlusion scores, and enhancing human-object interaction modeling with structured quantitative visibility information.

The dataset is derived from the SHHQ dataset [42] and is augmented with realistic occluding objects to replicate challenging real-world conditions. We simulate occlusions by overlaying common clutter objects (O) (two different version each of bushes, stopsigns, cars, suitcases, billboards, benches, balls, umbrellas, hats, trees) onto RGB images (I), applying a variety of geometric and photometric augmentations (See Figure 1). Each sample consists of (i) an RGB image X containing a person with synthetic occlusion and (ii) a ground-truth occlusion vector $A_p = [a_{p1}, a_{p2}, a_{p3}]$ denoting occlusion extent in head, torso, and legs.

For each body part $p_i \in P = \{\text{head, torso, legs}\}$, let n_{p_i} and n'_{p_i} denote the number of pixels before and after occlusion, respectively. The occlusion extent for that part is then

Algorithm 1 Creating the Occlusion and Finding the Ground-Truth

Inputs: Image(I), Silhouette(I'), Occluding object (O), augmentation operation (*operation*)
Output: Occluded image (X), Occluded silhouette (X'), Occlusion extent in each part as a list (A_p)

```

//Adding augmentation to the occluding object (O)
1:  $O \leftarrow \text{add\_augmentation}(O, \text{operation})$ 
//Getting the binary mask of the occluding object
2:  $O' \leftarrow \text{make\_binary}(O)$ 
//Getting a point on the body to paste the occlusion
3:  $\text{position} \leftarrow \text{random\_center}(I')$ 
//Getting the occluded image
4:  $X \leftarrow \text{superimpose}(I, O, \text{position})$ 
//Getting the occluded silhouette
5:  $X' \leftarrow \text{subtract}(I', O', \text{position})$ 
//Getting the ground-truth occlusion extent
6:  $A_p \leftarrow \text{get\_label}(I', X')$ 
7: return  $X, X', A_p$ 

```

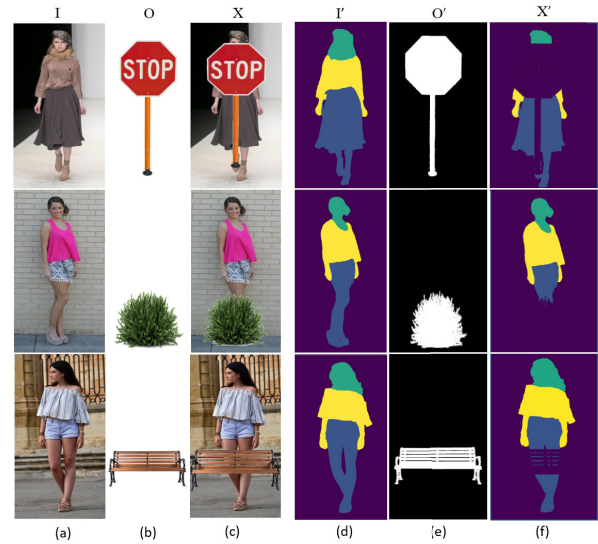


Fig. 1. (a) Original image (I). (b) Occluding object (O). (c) Occluded image (X). (d) Parsed original image (I'). (e) Binary mask of the occluding object (O'). (f) Parsed occluded image (X'). O undergoes random augmentations and is pasted onto I at a specific location, defined by the coordinates (x, y) , resulting in X . Subsequently O' undergoes identical augmentations as O and is subtracted at identical coordinates (x, y) from I' to produce X' .

given by

$$a_{p_i} = \frac{n_{p_i} - n'_{p_i}}{n_{p_i}} \quad (1)$$

yielding the ground-truth occlusion vector $A_p = [a_{p1}, a_{p2}, a_{p3}]$ for the image. The dataset generation follows Algorithm 1 and the corresponding process is visualised in Figure 1.

The final dataset comprises **1.2 million annotated image-label pairs**, covering 20 occluder types with two appearance variants each of bushes, stopsigns, cars, suitcases, billboards, benches, balls, umbrellas, hats and trees. Table II

provides a summary showing balanced coverage across body regions, with mean occlusion extents near 0.4 for all parts.

IV. TECHNICAL APPROACH-POANET

The proposed Parts-based Occlusion Aware Network (POANet) integrates convolutional feature extraction with knowledge transfer. The Occlusion Detection Network captures both the presence and spatial extent of occlusion across body regions, and when combined with POANet, this occlusion-aware information significantly enhances ReID performance. This improvement underscores the effectiveness of standard convolutional neural network architectures in RGB-based ReID tasks, achieving strong results without relying on complex attention mechanisms or computationally intensive covariance calculations. An overview of the technical approach is presented in Figure 2. In the following seven sections, we describe the details of these components in the overall REID architecture.

A. HiRes Parser for Human Body

We use the MIT licensed Human Parsing module [43] trained on the ATR dataset [44] to parse the images. The model trained on the ATR datasets has 18 classes namely ‘Background’, ‘Hat’, ‘Hair’, ‘Sunglasses’, ‘Upper-clothes’, ‘Skirt’, ‘Pants’, ‘Dress’, ‘Belt’, ‘Left-shoe’, ‘Right-shoe’, ‘Face’, ‘Left-leg’, ‘Right-leg’, ‘Left-arm’, ‘Right-arm’, ‘Bag’, ‘Scarf’. After parsing, since we know the value and the part each label maps to, we aggregate the labels to finally conglomerate the output into 3 parts namely *head*, *torso* and *legs*.

Fine-grained parts such as hands or feet offer low semantic discriminability, as their embeddings often capture noise, yielding high intra-class variance relative to inter-class variance and weakening ReID performance. Excessive partitioning also reduces effective sample size per part (N/k for N samples and k parts), leading to high variance estimation, overfitting, and poor representation for frequently occluded regions. In contrast, coarser divisions (head, torso, legs) are more consistently visible, semantically meaningful, and statistically reliable, providing stronger identity cues. Empirical studies (e.g., in PCB [32]) confirm that while finer uniform splits capture more detail, they suffer from misalignment, whereas coarse semantic parsing yields more robust and discriminative embeddings under occlusion.

Since most ReID datasets contain relatively small images, direct parsing using models trained on larger resolutions [44] is ineffective. To address this, we integrate a diffusion model-based super-resolution module, ResShift [41], which first super-resolves the input image to $img_{superres}$. This enhanced image is then parsed [44] to produce more accurate and sharper part-wise human body segmentation (See Supplementary material Figure colored1).

To align the parsed output with the original image dimensions, we first resize the parsed map to match the input resolution. A contour detection algorithm [45] is then applied to extract bounding boxes for each body part. Using these coordinates, we crop the head, torso, and leg regions directly

from the original RGB image, yielding four inputs: the original full image (processed by the GE module) and three part-specific crops (processed by the PAFE module). The complete parsing procedure is described in Algorithm 2, where the *process_image* function performs bounding box extraction and cropping.

Algorithm 2 Superresolving and Parsing an Image

Input: *Image*

Output: Parsed and Cropped Images

//Super-resolving the input image

1: $superres_image \leftarrow superresolute(image)$

//Parsing the super-resolved image

2: $parsed \leftarrow parse(superres_image)$

//Resizing the parsed image

3: $resized \leftarrow resize_image(parsed)$

//Processing the image to obtain cropped regions

4: $cropped_images \leftarrow process_image(image, resized)$

5: **return** $cropped_images$

To address cases where a body part is heavily occluded or insufficiently visible, we introduce a size-based filter: if the bounding box dimensions fall below $thresh = (40, 40)$, the corresponding crop is replaced with a black image. During training, these placeholder crops are excluded from loss computation by applying the softmax and triplet losses only to non-black images. This ensures that backpropagation prioritizes reliably visible regions, thereby reinforcing the model’s focus on non-occluded, discriminative features.

B. Occlusion Detector Network (ODN)

A novelty of our framework is the **Occlusion Detector Network (ODN)**, designed to extract explicit occlusion-aware information from pedestrian images. The motivation for this module stems from the observation that existing ReID systems often fail to account for the *anatomical location and extent of occlusion*, resulting in corrupted embeddings when key identity-relevant regions are obscured. By explicitly estimating occlusion for each body part, we enable downstream components to selectively emphasize visible, discriminative regions and suppress occluded areas, thereby improving retrieval robustness in challenging scenarios.

The ODN is implemented using a ResNet-50 backbone pretrained on ImageNet, denoted $R50_{OD}$. This backbone processes an input RGB image $I \in \mathbb{R}^{H \times W \times 3}$ to produce a deep feature map:

$$F_o = R50_{OD}(I), \quad F_o \in \mathbb{R}^{C \times H' \times W'}. \quad (2)$$

where C is the channel dimension after the third residual block, and $H' \times W'$ is the corresponding spatial resolution.

To regress the occlusion extent for each predefined body part $P = \{\text{head, torso, legs}\}$, F_o is passed through a 3×3 convolutional layer with 2048 output channels followed by a 2×2 max pooling operation:

$$Fp = \text{MaxPool}_{2 \times 2}(\text{Conv}_{3 \times 3}^{2048}(F_o)), \quad (3)$$

TABLE I
PRINCIPLES OF DIFFERENT METHODS FOR HANDLING OCCLUSION ALONG WITH DATASETS. THE COMMENTS COLUMNS
HIGHLIGHT THE PIVOTAL POINT OF AN APPROACH

Work	Principle	Datasets	Comments
Boshra, Bhanu [3], [4]	Mathematical modeling of occlusion in feature space	MSTAR public data domain [5]	Finds an upper bound and a good lower bound on prediction error.
Liu et al. [6], Yao et al. [7], Zhao et al. [8]	Transformer-based attention	Market-1501 [9], CUHK03 [10], DukeMTMC [11]	Does not explicitly model occlusion and is computationally complex.
Gupta et al. [12]	Occlusion detector trained on silhouettes to guide gait recognition	BRIAR, GREW [13]	No information about the occlusion in the RGB domain.
Li et al. [1]	Transformer based network with second order attention for occlusion	Market-1501, CUHK03	Uses attention to extract occlusion features; high computational complexity.
Chen et al. [14]	Occlusion modeling using a novel augmentation method	Market-1501, DukeMTMC	Provides feature-level occlusion without physical location; coarse granularity.
Su et al. [15], Zheng et al. [16], Zhang et al. [17]	External body feature driven person Re-Identification	Market-1501, CUHK03	Uses global pose information, lacks finer detail and explicit occlusion modeling.
Zhang et al. [18], Fan et al. [19], Ke et al. [20]	Occlusion modeling with RF-vision, 3D occupancy estimation, and bilayer segmentation	Custom benchmark datasets from ICCV 2023	Robust feature extraction under occlusion using advanced sensing, geometric estimation, and separation of occluder from occludee.
Dutta, Bhanu. (This work)	Occlusion network and parts based person Re-Identification	Market, CUHK03, DukeMTMC, MSMT17, Occluded ReID, Occ-Duke, P-Duke, PETHZ, SHHQ-occ	Integrates occlusion-aware features into a by-parts ReID model to guide attention adaptively based on occlusion.

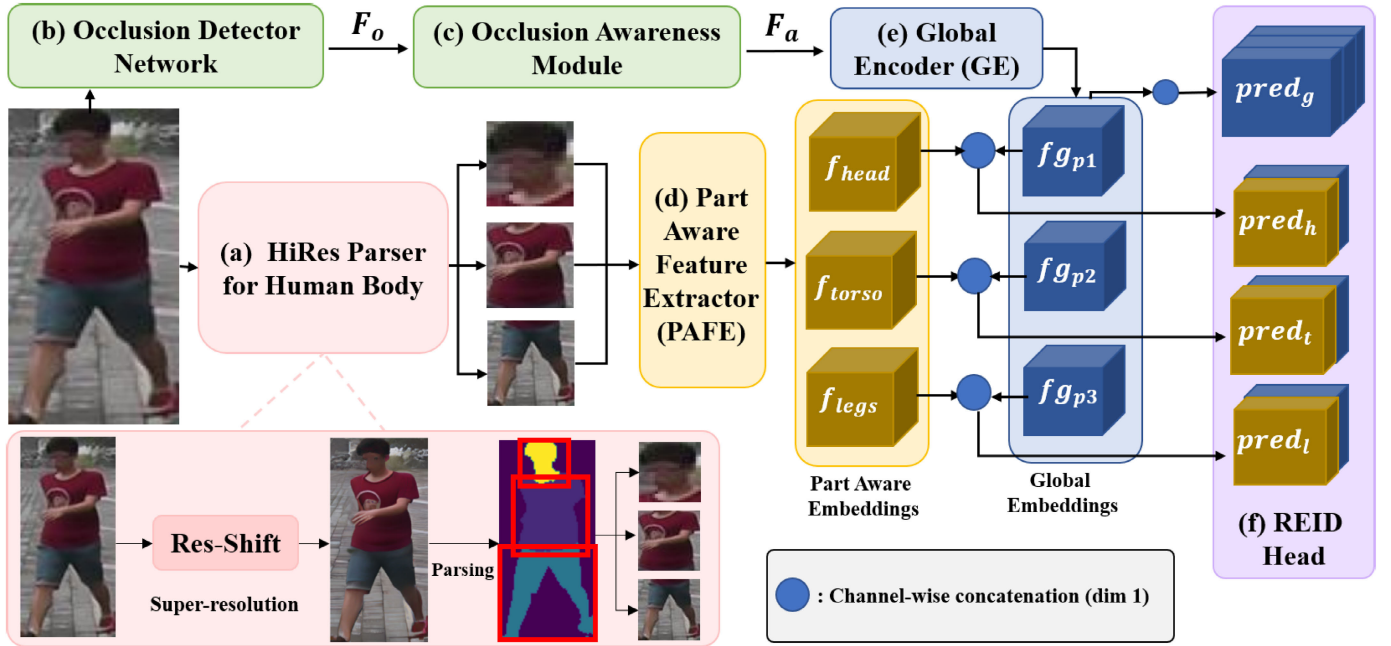


Fig. 2. Overview of the proposed technical framework. The architecture comprises six major components: (a) **HiRes Parser for Human Body**, which parses the input image into semantically segmented body parts after super-resolution using Res-Shift [41] (Section IV-A). (b) The **Occlusion Detector Network** (Section IV-B), which generates an occlusion feature map F_o . (c) The **Occlusion Awareness Module** (Section IV-C) that produces an occlusion aware global feature map F_a . (d) The **Part Aware Feature Extractor (PAFE)** module (Section IV-D), which extracts part-aware embeddings f_{head} , f_{torso} , and f_{legs} from the parsed body-part images. (e) The **Global Encoder (GE)** module (Section IV-E), which takes F_a as input and produces global embeddings fg_{pi} for $i = 1$ to 3. (f) The **ReID Head** (Section IV-F) which integrates information from both the GE and PAFE modules to produce the final identification embedding: $pred_{final}$.

The pooled feature map F_p is then flattened and passed vector:
through a fully connected layer with 1024 hidden units,
followed by a final regression layer producing a 3×1 output

$$\hat{A}_p = FC_3(FC_{1024}(\text{Flatten}(F_p))), \quad \hat{A}_p \in \mathbb{R}^{3 \times 1}, \quad (4)$$

TABLE II

THE MEAN, MEDIAN AND STANDARD DEVIATION OF THE OCCLUSION EXTENT IN EACH BODY PART IN OUR DATASET

	Head	Torso	Legs	Total
Mean	0.39	0.44	0.51	0.42
Median	0.37	0.35	0.39	0.41
Standard deviation	0.22	0.17	0.16	0.15

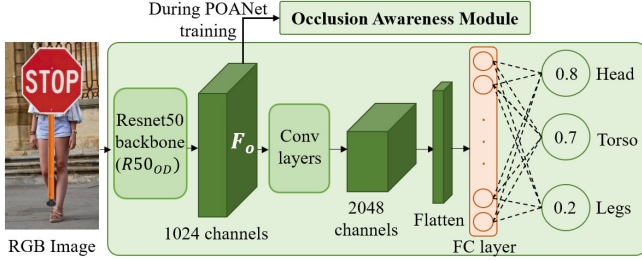


Fig. 3. Architecture of the Occlusion Detector Network. The values 0.8, 0.7, and 0.2 denote the predicted occlusion extents for the Head, Torso, and Legs, respectively, indicating high occlusion in the upper body and lower occlusion in the legs. Outputs are treated as independent regression values and are not passed through a softmax activation.

where $\hat{A}_p = [\hat{a}_{\text{head}}, \hat{a}_{\text{torso}}, \hat{a}_{\text{legs}}]^T$ contains the predicted occlusion extent for each body part, with each $\hat{a}_p \in [0, 1]$.

The network is trained using the Mean Absolute Error (L1) loss:

$$\mathcal{L}_{\text{OD}} = \frac{1}{3} \sum_{p \in P} |\hat{a}_p - a_p|, \quad (5)$$

where a_p denotes the ground-truth occlusion extent obtained from our SHHQ-Occ dataset (Section III). The Adam optimizer is used with momentum 0.9 and weight decay 5×10^{-4} .

The model is trained on 70% of the SHHQ-Occ dataset, validated on 20%, and tested on the remaining 10%. It achieves a Mean Absolute Error (MAE) of 0.02 on the held-out test set, demonstrating its ability to precisely estimate occlusion extent across all body parts.

Once trained, only the backbone output F_o is retained for integration into the ReID system via the Occlusion Awareness Module (Section IV-C). This integration ensures that the final ReID embeddings are informed by accurate occlusion information, a design choice that leads to substantial performance gains across all evaluated ReID benchmarks.

C. Occlusion Awareness Module (OAM)

A core novelty of our framework is the **Occlusion Awareness Module**, which transfers occlusion-relevant knowledge extracted using the pretrained ODN, into the ReID backbone. Its physical significance lies in enabling the global feature extractor to incorporate explicit occlusion information without requiring additional supervision or data augmentation during training. By fusing identity-discriminative features with fixed occlusion-sensitive features, the network gains the ability to selectively emphasize visible, discriminative regions and downweight occluded areas, thereby enhancing robustness in challenging real-world scenarios.

We employ two backbones with identical architectures—ResNet-50 truncated at the third residual block: (i) $R50_{\text{ReID}}$: the ReID backbone, fully trainable, responsible for learning identity-specific features. (ii) $R50_{\text{OD}}$: the ODN backbone, pretrained and frozen, providing occlusion-aware feature maps without weight updates. Given a full-body image input I , the feature maps from the two backbones are:

$$F_g = R50_{\text{ReID}}(I), F_o = R50_{\text{OD}}(I). \quad (6)$$

We concatenate these feature maps along the channel dimension to obtain the combined representation:

$$F_c = [F_g \parallel F_o], \quad (7)$$

where $\parallel$ denotes channel-wise concatenation. A 1×1 convolutional transformation $\gamma(\cdot)$ is then applied to fuse identity-aware and occlusion-aware channels into a unified occlusion-aware global feature map, F_a which serves as the occlusion-aware embedding used for downstream global feature extraction and retrieval:

$$F_a = \gamma([F_g \parallel F_o]), \quad (8)$$

D. Part Aware Feature Extractor (PAFE) Module

A key novelty of our ReID framework is the **Part Aware Feature Extractor (PAFE)** module, which explicitly computes feature embeddings for each human body region separately. The motivation for this design is twofold. First, by isolating and learning representations for the head, torso, and legs, the network preserves the *anatomical significance* of these regions, ensuring that the learned features are semantically aligned with the physical structure of the human body, in contrast to fine grained feature learning employed by existing techniques which has some inherent drawbacks (See Section V-D second point). Second, this separation forms the basis for *part-based focus* in later stages of the pipeline: when concatenated with the global representation and processed by an attention mechanism, the model can modulate the contribution of each part according to its visibility. For instance, if the head is heavily occluded, the concatenated representation can downweight f_{head} and place greater emphasis on less-occluded parts for robust retrieval.

Technically, the PAFE module processes each parsed body-part image I_{p_i} independently for $p_i \in P = \{\text{head, torso, legs}\}$ through an identical convolutional subnetwork $\Phi_{\text{PAFE}}(\cdot)$, consisting of five sequential convolutional blocks with channel progression $64 \rightarrow 128 \rightarrow 256 \rightarrow 1024 \rightarrow 2048$, each followed by ReLU activation (and appropriate pooling/stride operations for downsampling), yielding the final part-specific embedding:

$$f_{p_i} = \Phi_{\text{PAFE}}(I_{p_i}), \quad p_i \in P. \quad (9)$$

$f_{\text{head}}, f_{\text{torso}}, f_{\text{legs}}$ denotes the output feature maps for each part. Each of these f_{p_i} is passed through a part-specific fully connected transformation to obtain an m -dimensional classification embedding:

$$n_{p_i} = \mathbf{W}_{p_i} f_{p_i} + \mathbf{b}_{p_i}, \quad p_i \in P, \quad (10)$$

where m is the number of identities in the dataset, and $\mathbf{W}_{p_i}$ and $\mathbf{b}_{p_i}$ are learnable parameters specific to part p_i .

E. Global Encoder (GE) Module

While local part features provide fine-grained, spatially focused cues, our ablation studies confirm that *global representations consistently encapsulate critical contextual information* that significantly benefits retrieval performance. The Global Encoder (GE) module is designed to capture this holistic view of the person image in an occlusion-aware manner. By concatenating the global representation with each part-specific embedding from the PAFE module, the network can preserve anatomical structure while allowing global context to guide the weighting of local features. This fusion ensures that, for a given part, the downstream computation retains both fine-grained detail and global semantic consistency, enabling the network to dynamically emphasize less-occluded and more discriminative regions. The part-specific prediction embeddings are thus obtained as:

$$\text{pred}_{\text{part}} = [fg_{p_i} \parallel f_{\text{part}}], \quad \begin{matrix} \text{part} \in \{\text{h}, \text{t}, \text{l}\}, \\ i \in \{1, 2, 3\} \end{matrix} \quad (11)$$

where $\parallel$ denotes channel-wise concatenation, fg_{p_i} is the i -th global embedding from the GE module, f_{part} is the corresponding PAFE output.

In parallel, a purely global prediction embedding pred_g is formed by concatenating the three global embeddings from the GE module:

$$\text{pred}_g = [fg_{p_1} \parallel fg_{p_2} \parallel fg_{p_3}]. \quad (12)$$

The global prediction embedding pred_g is then passed through a fully connected layer to obtain the global classification embedding:

$$n_{\text{global}} = \mathbf{W}_g \text{pred}_g + \mathbf{b}_g, \quad (13)$$

where $\mathbf{W}_g$ and $\mathbf{b}_g$ are the learnable weight matrix and bias vector, respectively.

Implementation-wise, the occlusion-aware global feature map F_a is processed through three parallel ResNet50-based branches. The first branch applies the third and fourth residual layers followed by a 1×1 convolution to produce the 2048-dimensional embedding fg_{p_1} . The second and third branches add a downsampling step before the same ResNet50 layers, yielding fg_{p_2} and fg_{p_3} . Together, $\{fg_{p_1}, fg_{p_2}, fg_{p_3}\}$ form the global embeddings used in Eq. (11) to generate the final part-specific predictions $\{\text{pred}_h, \text{pred}_t, \text{pred}_l\}$.

F. ReID Head

The ReID Head integrates the outputs of the Global Encoder (GE) and the Part Aware Feature Extractor (PAFE) modules to produce the final discriminative embedding for retrieval. The motivation for this design is to combine the complementary strengths of global and part-specific representations while explicitly incorporating occlusion-awareness into the final decision. By fusing these features and refining them through a channel and spatial attention mechanism, the network learns to emphasize unoccluded, identity-relevant regions while downweighting occluded or noisy areas.

Implementation-wise, the ReID head receives the part-specific prediction embeddings (f_{part}) concatenated

with the global embeddings for the corresponding scale (fg_{p_i}) as $\text{pred}_{\text{part}}$ according to Equation 11.

The complete prediction tensor is then obtained by concatenating all prediction embeddings along the channel dimension and passing it through a Convolutional Block Attention Module (CBAM) [46], denoted $\psi(\cdot)$, which applies both channel and spatial attention:

$$\text{pred}_{\text{final}} = \text{CBAM}([\text{pred}_g \parallel \text{pred}_h \parallel \text{pred}_t \parallel \text{pred}_l]). \quad (14)$$

The output $\text{pred}_{\text{final}}$ constitutes the occlusion-aware, part-guided final embedding used for retrieval. The classification embeddings from PAFE (Equation 10) and GE (Equation 13) are concatenated to form a unified vector for classification loss calculation, which is then passed through a single fully connected layer $\varphi(\cdot)$ to obtain the final classification embedding:

$$n_{\text{final}} = \varphi([n_{\text{global}} \parallel n_{\text{head}} \parallel n_{\text{torso}} \parallel n_{\text{legs}}]). \quad (15)$$

The embedding n_{final} is used in the classification loss, as described in Section IV-G.

By leveraging the anatomical structure encoded in the PAFE outputs and the holistic context captured by the GE module, the ReID Head ensures that attention is directed toward unoccluded, identity-discriminative features while suppressing less reliable, occluded regions.

G. Loss Functions

To fully exploit the discriminative power of the learned representations, we jointly employ softmax loss for classification and triplet loss for metric learning. At the end of the pipeline, part-aware embeddings are concatenated with the global representation and refined using CBAM, enabling the part-based focus scheme introduced in our novelty. This design mitigates the noise sensitivity of local features and leverages global context to establish stronger decision boundaries.

For basic discriminative learning, we treat the identification task as a multi-class classification problem. For the i -th learned feature f_i , the softmax loss is defined as:

$$L_{\text{softmax}} = - \sum_{i=1}^N \log \frac{e^{W_{y_i}^T f_i + b_{y_i}}}{\sum_{k=1}^C e^{W_k^T f_i + b_k}}$$

where N is the batch size during training, and C is the number of classes in the training dataset. We apply the softmax loss to n_{final} which we obtain from Equation 15.

We utilize the batch-hard triplet loss, an enhanced version of the original semi-hard triplet loss. This loss function is formulated as follows:

$$L_{\text{triplet}} = - \sum_{i=1}^P \sum_{a=1}^K \left[\alpha + \max_{p=1 \dots K} \|f_a^{(i)} - f_p^{(i)}\|^2 - \min_{\substack{n=1 \dots K \\ j=1 \dots P \\ j \neq i}} \|f_a^{(i)} - f_n^{(j)}\|^2 \right]_+ \quad (16)$$

In our formulation, $f_a^{(i)}$, $f_p^{(i)}$, and $f_n^{(i)}$ represent the features extracted from the anchor, positive, and negative samples,

TABLE III
DATASETS USED FOR TRAINING AND EVALUATION

Dataset	Training		Evaluation		#Cam
	#ID	#Img	#ID	#Img	
Market1501 (M) [48]	751	12,936	750	19,281	6
DukeMTMC-ReID [11]	702	16,522	702	19,889	8
CUHK03 [10]	767	7,368	700	6,728	2
MSMT17 [49]	1041	30,248	3060	82,161	15
Occluded-ReID [50]	100	8,281	100	6,824	2
P-DukeMTMC-ReID [11]	650	12,001	649	12,142	8
PETHZ [51]	285	5,612	286	5,596	2
Occluded-Duke [52]	812	9,522	1,000	6,096	8

respectively. We employ the triplet loss on the feature embedding $\text{pred}_{\text{final}}$ that we get from Equation 14 to enhance ranking performance.

Positive and negative samples refer to pedestrians with the same or different identities as the anchor. The triplet candidates are constructed using the furthest positive and closest negative pairs within a mini-batch containing P selected identities and K images per identity. This improved version of the triplet loss enhances the robustness in metric learning and further improves performance.

V. EXPERIMENTAL RESULTS

A. Datasets and Protocols

We evaluate our method on eight standard and occlusion-focused ReID benchmarks, consistently outperforming and equalling state-of-the-art approaches (Details in Table III). These results validate the effectiveness of our ODN and PAFE modules in robustly improving retrieval accuracy under challenging conditions.

1) *Performance Metrics and Protocols*: We evaluate the performance of ReID methods by reporting the cumulative match characteristics (CMC) at rank 1 and 5, as well as the mean average precision (mAP).

Holistic Datasets have been setup in single-probe mode. **Occluded datasets** have been setup for experiments as per the protocol in [48] where we use occluded person images as probes and full-body person images as galleries. Half of the identities are randomly selected for training and the other half for testing. We run the experiments 10 times and report the mean results. The Error row below Tables IV and V reports the Standard Deviation (SD) of the 10 runs of the results.

B. Implementation and Parameters

Input images are resized to 384×128 , and both backbone and branch networks use ResNet-50 pretrained on ImageNet [71]. Training applies random horizontal flipping and processes four inputs: the global image plus cropped head, torso, and leg regions. Each mini-batch samples $P = 16$ identities with $K = 8$ images, optimized using SGD (momentum 0.9, weight decay 0.0005) under triplet loss. The learning rate starts at 0.0002 and decays to 10^{-3} and 10^{-4} at epochs 90 and 95 over 100 epochs. These hyperparameters are fixed for all dataset experiments.

The Occlusion Detector Network (ODN) is trained separately on our **SHHQ-occ** dataset and kept fixed during ReID training. The GE module is initialized with its pretrained weights and fine-tuned jointly with the PAFE module. All experiments are implemented in PyTorch on a single NVIDIA RTX 3080 (12GB VRAM), with full training on Market-1501 taking about 2 hours. The model has 44M tunable parameters and occupies 1.514 GB of memory before data loading. For fair comparison, all datasets are evaluated under identical settings, using the Jaccard-based re-ranking scheme [51] to boost retrieval accuracy.

C. Results

Our implementation of POANet, which explicitly incorporates body part information, demonstrates superior performance compared to existing feature map division methods [32], [58], [63], as evidenced by the results in Tables IV and V. Specifically, our model outperforms current convolutional neural network (CNN)-based ReID approaches on the CUHK and DukeMTMC datasets, and achieves performance comparable to state-of-the-art methods on the Market-1501 dataset (Table V). Furthermore, our evaluations on occlusion-focused benchmarks, including Occluded-ReID, P-ETHZ, and P-DukeMTMC-ReID (Table IV), highlight that our model not only surpasses the state-of-the-art CNN-based methods but also achieves substantial gains when integrated with the Occlusion Detector Network. These findings demonstrate that explicitly incorporating occlusion information enhances the model's awareness of occluded regions and enables more robust optimization.

Use of retrieval augmentation ([2], [51]) is largely prevalent in ReID methods. To ensure full transparency and methodologically fair comparisons, we add explicit notation and a dedicated column in Tables IV and V regarding the use of retrieval augmentation techniques in previous methods. For more detailed analysis of each cited work and the impact of retrieval augmentation, please refer to section 4 of our Supplementary Material.

D. Discussions on Performance Improvement

We accredit the improvement in performance to three major ideas in our approach:

E. Effect of ODN

Our model effectively addresses occlusion by explicitly focusing on the visible, non-occluded body parts. Specifically, we incorporate occlusion information extracted from the ResNet50 backbone of the Occlusion Detector Network, which enables the model to prioritize non-occluded regions within the feature maps. Figure 4 compares retrieval with and without the occlusion module across three probe scenarios using local embeddings (f_{head} , f_{torso} , f_{legs}) and the global embedding f_g . SSIM similarities between probe and gallery embeddings show that the module shifts focus toward unoccluded regions, such as the torso or head, enabling correct retrieval despite partial

TABLE IV

COMPARISON OF RESULTS ON THE OCCLUDED DATASETS. ALL EVALUATIONS ARE DONE IN A SINGLE PROBE MODE. BOLD RESULT IN EACH COLUMN IS THE BEST RESULT, WHILE UNDERLINED VALUES DENOTE THE SECOND-BEST. RK REFERS TO RE-RANKING [51]. THE (*) REPRESENTS TRAINING TIME AUGMENTATION WITH OIOU [2]

Methods	Venue	Retrieval Augmentation	Occluded ReID		Occluded Duke		P-ETHZ		P-DukeMTMC-ReID	
			Rank-1	Rank-5	Rank-1	Rank-5	Rank-1	Rank-5	Rank-1	Rank-5
SVDNet[54]	ICCV 2017	✓	63.1	85.1	-	-	52.2	78.9	43.5	63.4
AFPB[50]	ICME 2018	✗	68.1	88.3	-	-	-	-	-	-
Teacher-S[55]	ArXiv 2019	✗	73.7	92.9	-	-	-	-	-	-
REDA [56]	TIP 2021	✓	65.8	87.9	-	-	54.4	79.1	45.2	62.9
IGOAS[57]	TIP 2021	✗	81.1	91.6	-	-	-	-	-	-
NetVLAD-M[58]	TIFS 2022	✓	89.8	96.5	-	-	-	-	-	-
SAP[59]	AAAI 2023	✗	83.0	89.7	70.0	82.9	59.6	80.4	71.6	80.3
HCGA[60]	TIP 2023	✓	88.0	96.0	-	-	-	-	71.9	78.7
PRE[61]	TCSVT 2023	✗	-	-	68.3	86.7	-	-	-	-
SPT[2]	AAAI 2024	✓*	87.8	94.5	74.7	85.8	-	-	-	-
SPH[62]	TCSVT 2024	✓	82.6	89.1	71.9	82.4	64.7	80.3	72.1	81.2
FCFormer[63]	TMM 2024	✓	87.1	90.7	79.4	84.3	-	-	90.6	92.5
ETND[64]	TCSVT 2024	✗	90.5	95.6	68.1	77.7	69.3	85.7	85.4	89.8
POANet (Ours)	-	✗	89.9	96.3	77.2	85.3	68.9	84.8	84.5	88.7
POANet+ODN (Ours)	-	✗	<u>91.1</u>	<u>97.0</u>	<u>80.5</u>	<u>88.2</u>	<u>72.1</u>	<u>86.1</u>	86.8	90.4
POANet+ODN+RK(Ours)	-	✓	91.8	97.2	81.6	90.2	75.4	87.6	<u>88.3</u>	<u>90.8</u>
Error (SD)	-		0.05	0.06	0.05	0.06	0.04	0.03	0.04	0.05

TABLE V

COMPARISON OF RESULTS ON HOLISTIC DATASETS. THE METHODS ARE DIVIDED ON THE BASIS OF DIFFERENT CORE METHODOLOGIES USED I.E., CONVOLUTIONAL NEURAL NETWORK BASED, EXTERNAL CUE BASED, AND TRANSFORMER-BASED AND OUR METHODS IN THE ORDER FROM TOP TO BOTTOM. BOLD RESULT IN EACH COLUMN IS THE BEST RESULT, WHILE UNDERLINED VALUES DENOTE THE SECOND-BEST. RK REFERS TO RE-RANKING [51]. THE (*) REPRESENTS TRAINING TIME AUGMENTATION WITH OIOU [2]

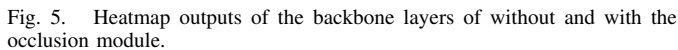
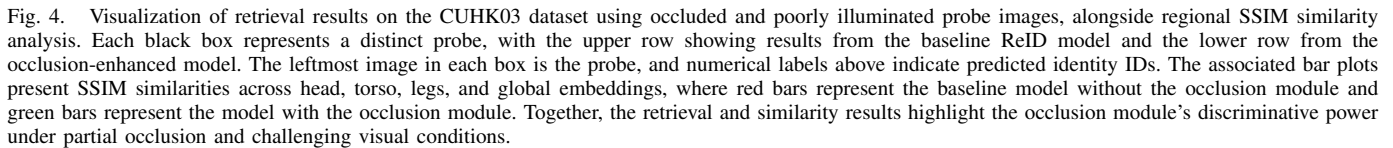
Methods	Venue	Retrieval Augmentation	MSMT17		DukeMTMC-ReID		Market-1501		CUHK03	
			Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP
PAN[16]	CVPR 2018	✓	-	-	71.6	51.5	-	-	36.3	34.0
HA-CNN[65]	CVPR 2018	✗	-	-	80.5	63.8	91.2	75.7	41.7	38.6
PCB+RPP[32]	ECCV 2018	✓	77.3	57.6	84.5	71.8	95.1	91.9	67.5	63.7
MGN [66]	MM 2018	✓	79.4	58.3	88.7	78.4	95.7	86.9	66.8	66.0
BPBReID[67]	WACV 2023	✗	80.1	59.6	89.6	78.3	95.1	87.0	-	-
3DInvarReID[68]	CVPR 2023	✗	80.8	59.1	-	-	95.1	87.9	-	-
PCB[32]	ECCV 2018	✗	75.8	56.0	83.3	69.2	92.3	77.4	63.7	57.5
GP-ReID[69]	TIP 2019	✓	-	-	85.2	72.8	92.2	81.2	-	-
Deep-Person[70]	CVPR 2018	✓	-	-	80.9	64.8	93.0	79.6	-	-
SDN [41]	Elsevier 2022	✓	-	-	90.9	81.0	96.2	95.1	83.7	84.5
HCGA [60]	TIP 2023	✓	85.7	72.8	-	-	95.2	88.4	76.9	73.2
MHSA-Net[71]	TNNLS 2023	✓	83.5	70.2	87.3	73.1	94.6	84.0	72.8	69.3
Soldier [72]	CVPR 2023	✓	90.7	77.1	-	-	<u>96.9</u>	<u>93.9</u>	-	-
PRE [61]	TCSVT 2023	✗	88.6	74.4	89.3	77.8	95.3	86.5	-	-
SCAT [73]	TII 2024	✗	-	-	89.3	79.8	95.1	88.0	-	-
SPT [2]	AAAI 2024	✓*	-	-	<u>91.1</u>	82.4	95.5	89.4	-	-
FCFormer [74]	TMM 2024	✓	-	-	<u>89.7</u>	78.8	95.0	86.8	-	-
ETND [64]	TCSVT 2024	✗	82.7	58.0	89.3	78.8	95.7	88.7	-	-
POANet (Ours)	-	✗	88.2	73.1	88.9	83.7	95.1	85.6	73.7	75.4
POANet+ODN (Ours)	-	✗	<u>90.8</u>	<u>77.1</u>	90.1	<u>86.5</u>	95.3	86.9	75.8	78.0
POANet+ODN+RK (Ours)	-	✓	91.2	77.5	93.6	87.9	97.5	88.7	<u>78.2</u>	<u>80.2</u>
Error (SD)	-		0.04	0.05	0.02	0.03	0.01	0.02	0.03	0.04

occlusion. This focus-shifting validates our self-supervised design, where the model learns to downweight occluded parts and emphasize visible cues, thereby improving alignment and retrieval performance.

As illustrated by the heatmap visualizations in Figure 5 ((a) Input images. (b) Heatmap output of network **without** the occlusion module. (c) Heatmap output of network **with** the occlusion module). This occlusion-aware design leads to a marked shift in focus compared to models without occlusion handling. In the presence of the Occlusion Detector module, the feature maps disregard the occluded clutter and instead emphasize the visible person regions. In contrast, the baseline model without the occlusion module lacks explicit occlusion

awareness and distributes attention indiscriminately across the entire image, including the occluded areas. Focusing on visible parts enhances accuracy, enabling the occlusion-aware backbone to extract discriminative features under occlusion.

Table IX reports results of training the ODN with varying numbers of occluder types (5, 10, 15, and 20). As occluder diversity increases, MAE steadily decreases (0.16 to 0.02) while ReID mAP improves across Duke, MSMT17, Occluded-ReID, and Occluded Duke. These results validate two key points: (i) the SHHQ-occ dataset provides a comprehensive foundation for learning occlusion-aware features, and (ii) the ODN generalizes effectively to unseen occluders, with robustness enhanced by exposure to greater occlusion



To further verify the effectiveness of the proposed Occlusion Detector Network (ODN), we integrated it into two standard ReID baselines, PCB and MGN, by attaching it through the Occlusion Awareness Module (OAM) and retraining under identical settings across all benchmark datasets. As shown in Table VI, the ODN consistently improves both mAP and Rank-1 accuracy, demonstrating that it provides complementary occlusion-aware features absent in the baselines. These results confirm that the ODN is not only effective within our framework but is also generalizable and easily adaptable to other ReID architectures, thereby validating our novelty of

1) *Effect of PAFE*: Notably, we introduce **PAFE** that significantly improves body-part identification and matching. Unlike existing approaches that divide feature maps into fixed horizontal strips [32], [63], our method preserves anatomically consistent representations for each region, avoiding the *semantic misalignment* caused when the same strip index corresponds to different body parts across images. For instance, a strip may capture legs in one image but torso in another, leading to noisy features. In contrast, our **PAFE module** explicitly models each body part (head, torso, legs), capturing fine-grained, anatomically meaningful details such as hair texture or clothing patterns. Crucially, PAFE allows the network to *dynamically shift focus*, downweighting occluded regions and emphasizing unoccluded ones, thereby enhancing robustness and achieving more accurate retrieval under occlusion.

Table VII presents an ablation study evaluating the impact of various embedding configurations on retrieval performance across the six ReID benchmarks. Overall, the global embedding ($pred_g$) consistently outperforms the individual part-based embeddings ($pred_h$, $pred_t$, $pred_l$), underscoring the effectiveness of holistic representations. However, we observe that integrating part-based embeddings alongside the global embedding leads to consistently improved performance across all datasets, highlighting the value of explicitly modeling localized body regions. Notably, in datasets such as Market1501 and DukeMTMC, certain part embeddings—particularly combinations like head+torso—achieve higher scores than the global embedding alone because it captures the most distinctive and less frequently occluded features for person identification, offering a richer and more robust representation than legs or individual parts alone. These results demonstrate the complementary nature of localized and global

TABLE VI

RETRIEVAL PERFORMANCE OF BASELINE METHODS (PCB AND MGN) AND THEIR ODN-AUGMENTED VARIANTS ACROSS EIGHT BENCHMARK DATASETS. EACH ENTRY REPORTS mAP AND RANK-1 ACCURACY. RELATIVE GAIN (%) IS COMPUTED AS $((X - \text{BASELINE})/\text{BASELINE}) \times 100$, (WHERE X IS PCB+ODN OR MGN+ODN) WITH POSITIVE IMPROVEMENTS MARKED BY $\uparrow$

Method	Market1501		CUHK03		DukeMTMC		MSMT17		Occluded-ReID		P-DukeMTMC		PETHZ		Occluded Duke	
	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1
PCB	77.4	92.3	57.5	63.7	69.2	83.3	56.0	75.8	70.6	72.1	43.4	45.5	61.8	54.1	58.0	66.3
PCB+ODN	78.3	93.1	59.2	65.6	72.6	85.7	59.4	77.8	72.1	73.8	45.2	48.7	63.9	58.1	60.0	68.8
Relative Gain (%)	1.16 $\uparrow$	0.87 $\uparrow$	2.96 $\uparrow$	2.98 $\uparrow$	4.91 $\uparrow$	2.88 $\uparrow$	6.07 $\uparrow$	2.64 $\uparrow$	2.13 $\uparrow$	2.36 $\uparrow$	4.15 $\uparrow$	7.03 $\uparrow$	3.40 $\uparrow$	7.39 $\uparrow$	3.45 $\uparrow$	3.77 $\uparrow$
MGN	86.9	95.7	66.0	66.8	78.4	88.7	58.3	79.4	72.4	74.9	46.6	48.0	63.2	57.5	61.7	67.9
MGN+ODN	87.2	96.1	67.8	79.9	80.3	89.6	59.1	80.7	75.2	77.4	49.2	51.4	65.6	59.4	63.8	69.9
Relative Gain (%)	0.35 $\uparrow$	0.42 $\uparrow$	2.73 $\uparrow$	19.61 $\uparrow$	2.42 $\uparrow$	1.01 $\uparrow$	1.37 $\uparrow$	1.64 $\uparrow$	3.87 $\uparrow$	3.34 $\uparrow$	5.58 $\uparrow$	7.08 $\uparrow$	3.80 $\uparrow$	3.30 $\uparrow$	3.40 $\uparrow$	2.94 $\uparrow$

TABLE VII

RETRIEVAL PERFORMANCE OF OUR MODEL USING DIFFERENT EMBEDDING CONFIGURATIONS ACROSS EIGHT PERSON REID DATASETS. ROWS INDICATE THE EMBEDDING COMBINATIONS (PART-BASED OR GLOBAL) USED DURING INFERENCE. BOLD VALUES INDICATE THE BEST RESULT IN EACH COLUMN, WHILE UNDERLINED VALUES DENOTE THE SECOND-BEST. ODN IS USED IN EACH

Embedding	Market1501		DukeMTMC		CUHK03		MSMT17		Occluded-ReID		P-DukeMTMC		PETHZ		Occluded Duke	
	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1	mAP	R1
Head ($pred_h$)	80.2	85.5	78.1	82.6	61.4	60.9	62.1	78.9	55.4	57.8	61.8	66.2	50.8	47.1	52.1	60.4
Torso ($pred_t$)	77.3	79.9	65.3	70.2	63.2	59.7	64.8	79.5	72.3	70.9	57.4	59.9	62.9	59.4	58.3	66.8
Legs ($pred_l$)	36.4	33.1	29.1	31.7	34.8	31.5	37.1	42.6	29.5	28.3	33.6	35.3	32.6	30.5	31.6	33.5
Head + Torso	83.5	84.2	80.3	84.7	65.9	62.1	75.1	88.4	77.8	78.1	81.1	81.9	64.8	61.5	68.3	77.4
Head + Legs	81.2	83.9	78.6	81.5	63.8	60.3	68.3	77.9	73.4	74.2	79.3	80.7	60.6	58.1	59.7	64.1
Legs + Torso	79.8	81.4	75.3	77.7	60.2	59.9	56.8	71.0	67.3	67.8	73.2	74.8	57.8	55.3	57.9	62.8
Global ($pred_g$)	82.6	86.3	81.5	83.8	<u>74.9</u>	<u>72.1</u>	<u>75.6</u>	87.2	<u>85.4</u>	<u>86.2</u>	<u>83.7</u>	<u>84.3</u>	<u>77.5</u>	<u>74.8</u>	<u>69.4</u>	<u>79.2</u>
All ($pred_{final}$)	88.7	97.5	87.9	93.6	80.2	78.2	77.5	91.2	90.5	91.8	85.3	88.3	81.4	75.4	73.6	81.6

TABLE VIII

COMPARISON OF RESULTS ACROSS EIGHT DATASETS USING TWO TRAINING PROTOCOLS. **PROTOCOL 1**: END-TO-END TRAINING OF THE ENTIRE MODEL. **PROTOCOL 2**: TRAINING ONLY GE FOR 75 EPOCHS FOLLOWED BY FINE-TUNING WITH THE PAFE MODULE FOR 25 EPOCHS

Protocol	CUHK03		Market1501		DukeMTMC		MSMT17		Occluded-ReID		P-DukeMTMC		PETHZ		Occluded Duke	
	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP
Protocol 1	71.8	72.5	91.7	83.3	88.4	83.7	87.8	75.3	85.4	84.4	83.1	80.6	69.9	75.7	78.4	70.8
Protocol 2	78.2	80.2	97.5	88.7	93.6	87.9	91.2	77.5	91.8	90.5	88.3	85.3	75.4	81.4	81.6	73.6

TABLE IX

PERFORMANCE OF THE ODN TRAINED WITH VARYING NUMBERS OF OCCLUDERS ON SHHQ-Occ. WE REPORT OCCLUSION PREDICTION ACCURACY (MAE $\downarrow$) AND REID PERFORMANCE (mAP $\uparrow$) ACROSS BENCHMARK DATASETS

Training Occluders	MAE (SHHQ-occ)	mAP (Duke)	mAP (MSMT17)	mAP (Occluded-ReID)	mAP (Occluded Duke)
5 Occluders	0.16	84.2	74.9	87.4	69.4
10 Occluders	0.09	85.9	76.8	88.5	71.6
15 Occluders	0.05	87.2	77.3	89.2	72.9
20 Occluders	0.02	87.9	77.5	90.5	73.6

cues, and validate the importance of human body parsing in enhancing robustness, especially under partial occlusion or viewpoint variation.

2) *Local Parts Contain Finer Details*: We argue that local body parts contain finer-grained details, whereas the global image provides only coarser, holistic information. To leverage this distinction, we run ablation with two training protocols. We define **Protocol 1**: Training the entire model end to end together. and **Protocol 2**: Training the Global model first for 75 epochs and then fine tuning with the PAFE module for 25 epochs. Protocol 2 leads to enhanced performance, achieving higher rank-1 accuracy and mean average precision (mAP) scores, as demonstrated in Table VIII, when compared to Protocol 1. Notably, the jointly trained model underperforms relative to the fine-tuned approach, underscoring that local part information, due to its finer granularity, is better exploited during the fine-tuning stage rather than when blended from the

outset with coarser global features. This finding highlights the importance of progressive refinement in training, allowing the network to effectively capture complementary coarse and fine cues.

F. Discussion on the Foundational Questions Related With Occlusion

1) *How Effective is Occlusion Modeling For Real-World Usage?*: In Section III we presented our method to model and calculate the extent and location of occlusion. Since our occlusion dataset that we created consisted of high quality images, we resized and blurred the images to resemble the datasets we use for the ReID task. As explained in IV-B, our use of real-world images of benches, stop signs, bushes, etc. for the occlusion modeling seeks to replicate occlusion cases better than other works [12], [72] that use silhouette representation and random erasing for occlusion modeling.

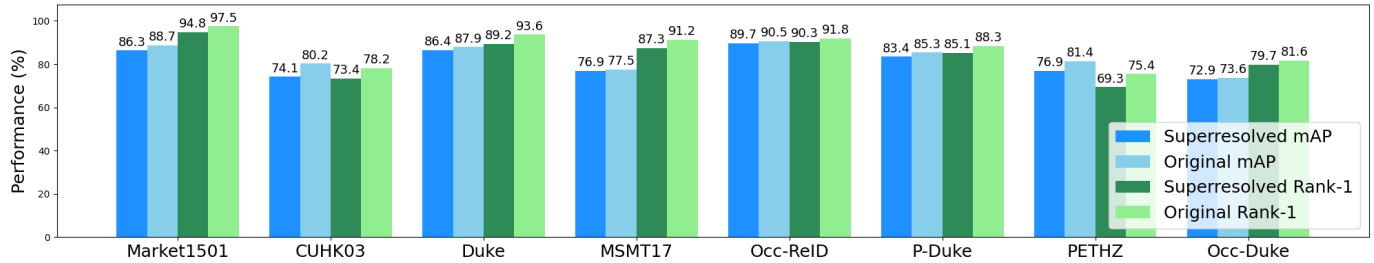


Fig. 6. Comparison of mAP and Rank-1 accuracy across six benchmark datasets between models using Super-resolved image parts and original images. The results consistently show that using original images yields superior Rank-1 performance compared to using Super-resolved parts, highlighting the limitations of Super-resolution for enhancing identity discrimination in person re-identification tasks.



Fig. 7. (a) Original images from the Market-1501. (b) Super-resolved images. Although the Super-resolved parts are clearer and have a higher resolution axiomatically, they hallucinate on facial features, thereby, misguiding the ReID pipeline to output wrong results.

2) *Why Not Use Attention Mechanisms to Get Better Results?*: While attention-based mechanisms have shown effectiveness for occlusion modeling in person ReID [1], [6], they incur high computational cost due to long-range dependency modeling. Li et al. [1] achieved state-of-the-art results on Occluded-ReID, Market-1501, and CUHK using two specialized attention heads, but required Nvidia RTX 4090 GPUs and did not report training time. In contrast, our model achieves superior performance with lightweight CNN feature extraction and part-aware occlusion modeling, training in just 2 hours on two RTX 3080 GPUs (1 minute/epoch on Market-1501). Moreover, Li et al.'s model contains an estimated **100–150M** parameters, whereas ours has only **44M**, yielding $\sim 0.35\times$ the parameter count and a **2.84x** speedup in training and inference. These results highlight the scalability and efficiency of our CNN-based approach, demonstrating that CNNs can effectively model occlusion-aware dependencies while offering a more practical alternative to transformer-based designs.

3) *Why are We Not Using the Super-Resolved Parts for the ReID Matching?*: While super-resolution improves sharpness and edge detail, it often introduces artefacts that degrade semantic and identity-preserving information. As shown in Figure 7, super-resolution can hallucinate facial features, leading to identity loss. We quantified this effect by resizing both the original and super-resolved images to (128, 384) and computing the SSIM similarity, we obtain a low score of **0.42**, that indicates significant dissimilarity. Figure 6 further confirms that using super-resolved parts yields worse CMC and mAP scores compared to original images. Thus, while super-resolved images aid parsing, we retain the original images for identity matching to preserve critical semantic cues.

VI. CONCLUSION

In this paper, we presented the **Parts-based Occlusion Aware Network (POANet)**, a convolutional framework that tackled the long-standing challenge of occlusion in person ReID. Central to our design are two novelties: (a) the **Occlusion Detector Network (ODN)** along with the **Occlusion Awareness Module (OAM)**, which explicitly modeled occlusion in the feature space and integrated occlusion information into global and local embeddings without reliance on external pose cues, and (b) the **Part Aware Feature Extractor (PAFE)**, which learned anatomically consistent part-based embeddings and dynamically shifted focus from occluded to visible regions. Together, these modules enabled robust feature learning under occlusion and achieved state-of-the-art results among convolutional models, while remaining competitive with transformer-based methods using only **35%** of their parameters. We further introduced the large-scale **SHHQ-occ dataset** (1.2M images) with precise part-level occlusion annotations, which proved critical for training ODN and validating its effectiveness across diverse occlusion scenarios. We also showed that attaching ODN and OAM to established backbones such as PCB and MGN led to consistent performance improvements, underscoring the modularity and broad applicability of our approach. Beyond person ReID, the principles established here—explicit occlusion modeling and anatomically grounded feature extraction—offer broad potential for advancing related tasks such as gait recognition, pose estimation, and segmentation, thereby setting a foundation for more robust and efficient vision systems in real-world conditions.

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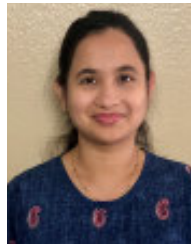
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